

Lecture 2: Eigenvalues & Spectral Mapping

Core Definitions

In this session, we analyze how linear transformations scale specific vector subspaces.

Definition 2.1 : Eigenvalues and Eigenvectors

Let $T : V \rightarrow V$ be a linear operator on a vector space V over a field F . A non-zero vector $v \in V$ is called an **eigenvector** of T if there exists a scalar $\lambda \in F$ such that:

$$T(v) = \lambda v$$

The scalar λ is called the **eigenvalue** corresponding to the eigenvector v .

Fundamental Theorems

Theorem 2.2 : Linear Independence of Eigenvectors

Let $T : V \rightarrow V$ be a linear operator. If $v_1, v_2, \dots, v_m$ are eigenvectors corresponding to **distinct** eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$, then the set $\{v_1, v_2, \dots, v_m\}$ is linearly independent.

Homework Practice

Exercise 2.3 : Characteristic Polynomial Evaluation

Consider the 2×2 matrix A representing a linear transformation on $\mathbb{R}^2$:

$$A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$$

1. Compute the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.
2. Find the distinct eigenvalues λ_1 and λ_2 by solving:

$$\lambda^2 - 7\lambda + 10 = 0$$